SESSION 7

Final Exam Review

Everything the exam can ask, and the twelve drills that cover it.

13 November 2026 · three hours, two breaks

Exam: Friday 27 November · one hour · no notes, no devices

How today works

Three hours, twelve drills, one mock paper

0:00 – 0:15	What the exam covers, and what it does not
0:15 – 0:30	The course in one picture, and the four equations
0:30 – 0:55	Part 2 — toolkit rapid-fire: drills 1 to 5
0:55 – 1:05	*Break*
1:05 – 1:35	Part 3 — balance of payments and the external position: drills 6 to 8
1:35 – 2:05	Part 4 — the intertemporal model: drills 9 and 10
2:05 – 2:15	*Break*
2:15 – 2:45	Part 5 — the three shocks: drills 11 and 12
2:45 – 3:00	The mock paper, and what to do in the next two weeks

Bring a pen. You will be writing for most of three hours — that is the point.

What the exam covers

Six teaching sessions, three chapters, one toolkit

IN SCOPE

- Session 0 — all five tools: the identities, stocks and flows, real versus nominal, discounting and present value, exchange-rate conventions.
- Chapter 1 — balance-of-payments accounting and the components of the current account.
- Chapter 2 — the NIIP, valuation changes, and current-account sustainability.
- Chapter 3 — the endowment economy in full: budget constraint, Euler equation, free capital mobility, log preferences, and the three shocks.

NOT IN SCOPE

- The production economy and the investment schedule (Ch. 5).
- Fiscal policy, twin deficits, Ricardian equivalence (Ch. 8).
- The TNT model and the theory of the real exchange rate (Ch. 9–10).
- Covered and uncovered interest parity, capital controls (Ch. 11–12).
- Currency pegs, balance-of-payments crises, optimum-currency-area theory (Ch. 14–15).

One exception worth knowing: the real exchange rate definition from Session 0 is fair game. The model of it is not.

1

The course in one picture

Where every question comes from

15 minutes

The whole course, in one line of logic

Four steps. Every exam question lives on one of them.

AN IDENTITY

$$CA = S - I$$

True by construction. No theory, no assumptions, nothing to disagree with.

A STOCK

$$NIIP_1 = NIIP_0 + CA + \text{valuation}$$

The flow accumulates into a position. A deficit is not a debt.

A MODEL

$$C_1 = \tfrac{1}{2}\,\bar{Y}$$

Now we add behaviour: households smooth consumption over time.

A PREDICTION

$$\Delta CA_1 = \tfrac{1}{2}\,\Delta Q_1$$

Which lets us say what the current account should do, and check it.

The exam walks left to right. Nothing in it comes from outside these four boxes.

If you are ever lost in a question, work out which box you are standing in. That alone usually tells you which equation to write down.

The four equations to know cold

If you can write these from memory you can attempt every question on the paper

1

$CA = S - I \quad \text{and} \quad NX = Y - A$

The identity. Also CA = TB + net income from abroad — the two differ by the interest on past positions, and the exam will test that you know it.

2

$NIIP_1 = NIIP_0 + CA_1 + \text{valuation}$

The stock-flow relation. Cutting a deficit slows the debt; it does not repay it.

3

$C_1 + C_2/(1+r^*) = (1+r_0)B_0 + Q_1 + Q_2/(1+r^*) \equiv \bar{Y}$

The intertemporal budget constraint. Everything on the right is lifetime wealth.

4

$C_1 = \tfrac{1}{2}\,\bar{Y}$

Log preferences with $\beta = 1$. Consume half of lifetime wealth in the first half of life. Then $TB_1 = Q_1 - C_1$ and $CA_1 = TB_1 + r_0 B_0$.

Write all four out, from memory, before you look at the paper on 27 November. Three minutes well spent.

2

Toolkit rapid-fire

Drills 1 to 5 — the cheapest marks on the paper

25 minutes

Drills 1, 2 and 3

Pens down when I say. Work alone first.

1

A country in one year, in billions of euro: output Y = 100, consumption C = 62, investment I = 24, government G = 20. There is no net income from abroad.

Find net exports two ways — from the spending side and from the saving side — and state the current account.

4 minutes

2

Same country, same year. Now you learn it also pays 3 of net investment income to foreigners, on debt it ran up in the past.

What happens to the trade balance? What happens to the current account? What is national saving now?

4 minutes

3

A country's net international investment position is −50% of GDP. It runs current-account deficits of 5% of GDP for three years, then cuts the deficit to 2% and holds it. Ignore valuation changes and assume GDP is flat.

Write the position at the end of each of the six years. Then answer: has the austerity worked?

5 minutes

Drills 1, 2 and 3 — answers

Two routes to one number, and a stock that keeps climbing

1

Absorption A = 62 + 24 + 20 = 106
Spending side: NX = Y − A = −6
Saving side: S = Y − C − G = 18
so S − I = 18 − 24 = −6

Both routes give −6, and CA = −6.

2

The trade balance does not move. It is a flow of goods, and no goods changed hands.

CA = TB + net income = −6 − 3 = −9
GNI = 100 − 3 = 97
S = 97 − 62 − 20 = 15, S − I = −9 ✓

3

Year	CA	NIIP
1	−5	**−55**
2	−5	**−60**
3	−5	**−65**
4	−2	**−67**
5	−2	**−69**
6	−2	**−71**

Drill 2 in one sentence: the trade balance is about goods, the current account is about goods plus what you owe on yesterday's borrowing.

Drill 3 — has the austerity worked?

It depends entirely on which number you were promised. The flow improved by three fifths — that is real. The stock is 21 points of GDP worse than when they started, and still rising every year. Both sentences are true, and a politician can pick either one. Closing a deficit slows the accumulation of debt; only a surplus reverses it.

Exam habit: whenever a question gives you a deficit and a debt in the same breath, write down which is the flow and which is the stock before you compute anything.

Drills 4 and 5

Real versus nominal, and which way round the price is quoted

4

A country in a currency crisis has a nominal interest rate of 60% and inflation of 55%.

(a) Use the approximation $r \approx i - \pi$ to get the real interest rate.
(b) Now use the exact formula $1 + r = (1 + i) / (1 + \pi)$.
(c) Which is right, and why does the gap matter here but not in a normal euro-area year?

5 minutes

5

The ECB quotes the euro-dollar rate as 1 EUR = 1.08 USD in January, and 1 EUR = 1.16 USD in June.

(a) Which way did the euro move, and what is the word for it?
(b) A US source quotes the same market the other way round. Write both numbers as they would appear, and say which direction the number moves.
(c) Is ‘devaluation’ the right word for any of this?

6 minutes

Drills 4 and 5 — answers

When the shortcut breaks, and four words that are not synonyms

4

Approximation	**r ≈ 60 − 55 = 5.00%**
Exact	**1.60 / 1.55 − 1 = 3.23%**
The shortcut overstates by	**1.77 percentage points**

The exact formula is always right. The approximation drops a term that is negligible when i and π are small and is not negligible here.

5

ECB convention, USD per EUR	**1.08 → 1.16 rises**
US convention, EUR per USD	**0.926 → 0.862 falls**
The euro has	**appreciated**

One event, two series, opposite directions, both correct. Always read the denominator before you read the chart.

Drill 5(c) — no. And this distinction is examinable.

Appreciation	the currency buys more	the market	*floating*
Depreciation	the currency buys less	the market	*floating*
Revaluation	the parity is moved up	the authorities	*fixed*
Devaluation	the parity is moved down	the authorities	*fixed*

Inside the euro area the nominal rate between members is 1, permanently. So ‘devaluation’ is not available, and any real adjustment has to run through prices and wages.

10:00

Part 3 next: the balance of payments and the external position. Drills 6, 7 and 8.

3

The balance of payments and the external position

Drills 6 to 8 — Chapters 1 and 2

30 minutes

What is actually in the current account

Four components. The exam expects you to name all four.

Trade balance

Exports minus imports of goods and services. The biggest component for most countries, and the one headlines mean when they say ‘trade deficit’.

Net investment income

Interest, dividends and reinvested earnings received from abroad, minus the same paid to foreigners. This is the r_0B_0 term in the model.

Net compensation of employees

Wages earned by residents working abroad minus wages paid to non-resident workers. Small for most countries, large for cross-border commuter regions.

Net unilateral transfers

Money given with nothing received in return: remittances, foreign aid, EU structural funds. A pure gift, recorded as a current transaction.

The one rule for signs: anything that gives a resident a claim on a foreigner is a credit. Anything that gives a foreigner a claim on a resident is a debit.

Drill 6

Six transactions. Which account, and which sign?

For each one, name the account — and if it is in the current account, name which of the four components — then say whether it is a credit or a debit for the country named.

(a) A Spanish engineering firm ships machinery to a buyer in Brazil.

(b) A German household receives a dividend on shares it holds in a US company.

(c) A resident of Poland commutes daily across the border to a job in Germany and is paid there.

(d) An Italian family sends €2,000 to relatives living outside the EU.

(e) A French pension fund buys a newly issued German government bond.

(f) Romania receives a payment from the EU structural funds.

8 minutes

One of these six is not in the current account at all. Finding it is most of the question.

Drill 6 — answers

Five current-account entries and one impostor

(a)	**Trade balance**	Credit for Spain	*Goods leave, a claim on a foreigner arrives.*
(b)	**Net investment income**	Credit for Germany	*A return on an asset already owned abroad.*
(c)	**Compensation of employees**	Credit for Poland	*The worker is resident in Poland; the wage is income from abroad.*
(d)	**Unilateral transfers**	Debit for Italy	*Money out, nothing received in return.*
(e)	**Financial account — not the current account**	—	*Buying an asset is not income. The return on it, later, will be.*
(f)	**Unilateral transfers**	Credit for Romania	*A grant is a current transfer, not borrowing.*

Buying an asset is a financial-account entry. Earning on an asset is a current-account entry. The exam will test that line.

The stock-flow relation, and the term that ruins it

Chapter 2. A country's external position can worsen without it borrowing a cent.

$$NIIP_1 = NIIP_0 + CA_1 + \text{valuation changes}$$

$NIIP_0$

Where you started

What the country owned abroad minus what it owed, at the start of the year. A stock, with a date.

CA_1

What you did this year

A surplus adds to the position, a deficit subtracts. This is the part a government can act on.

Valuation

What happened to you

Prices and exchange rates moved. Nobody borrowed, nobody lent, and the position changed anyway.

Why the third term is not a footnote

A country whose foreign assets are in dollars and whose liabilities are in euro gains when the dollar rises, and loses when it falls — with no transaction of any kind. For countries with large gross positions, the valuation term can be bigger in a single year than a decade of current-account balances. This is why the current account alone does not tell you how a country's external position is evolving.

The current account is a decision. The valuation change is weather.

Drill 7

A year in the life of an external position

7

A country begins the year with a net international investment position of −500 billion euro. During the year:

- it runs a current-account deficit of 30 billion;
- the foreign-currency assets it holds abroad gain 12 billion in euro terms as the dollar rises;
- foreign investors' holdings of its domestic equities rise in value by 25 billion as its stock market rallies.

Find the position at the end of the year. Then say how much of the deterioration the country actually chose.

7 minutes

Careful with the third bullet. A rise in the value of what foreigners own of you is not good news for your net position.

Answer

Valuation on assets	**+12**
Valuation on liabilities	**−25 (a liability that grows worsens the net position)**
Net valuation change	**−13**
$NIIP_1$ = −500 − 30 − 13	**−543**

The position worsened by 43. Only 30 of that was borrowing — the other 13 happened to them.

What sustainability actually requires

Not a balanced current account. Something much weaker than that.

Start from the intertemporal budget constraint and move consumption to the left:

The constraint	$C_1 + C_2/(1+r^*) = (1+r_0)B_0 + Q_1 + Q_2/(1+r^*)$
Collect the Q and C terms	$(Q_1 - C_1) + (Q_2 - C_2)/(1+r^*) = -(1+r_0)B_0$
And those brackets are trade balances	$TB_1 + TB_2/(1+r^*) = -(1+r_0)B_0$

What it says

The present value of a country's current and future trade balances has to equal what it owes today. A net debtor must eventually run trade surpluses large enough, in present value, to cover its debt.

What it does not say

It does not say the current account must balance, this year or any year. It does not forbid deficits, or long runs of them. It rules out exactly one thing: a plan that never repays anything at all.

This is the no-Ponzi condition doing its work: you cannot pay interest by borrowing more, forever. That is the whole restriction.

Drill 8 ★

Bonus credit only. Attempt it if the rest came easily.

8 A country's net external position is −80% of GDP. Nominal GDP grows at 5% a year. Valuation changes are zero.

What current-account balance, as a share of GDP, keeps the ratio at −80% forever? Then: what happens to that number if growth falls to 1%?

8 minutes

Setting it up

Write b for NIIP as a share of GDP and ca for the current account as a share of GDP. In levels, $B_1 = B_0 + CA_1$. Divide through by this year's GDP, which is (1+g) times last year's:

$b_1 = b_0/(1+g) + ca_1$

Set $b_1 = b_0 = b$ and solve:

The answer

$$ca = b \times g / (1+g)$$

At g = 5%	**ca = −0.80 × 0.05/1.05 = −3.81%**
At g = 1%	**ca = −0.80 × 0.01/1.01 = −0.79%**

Read what just happened

Growth fell by four percentage points and the deficit this country can run forever fell from 3.8% of GDP to 0.8%. Nothing about its borrowing, its exports or its policy changed. A deficit that was comfortably sustainable became unsustainable because the denominator stopped growing — which is close to a one-line account of what happened to the euro-area periphery after 2008.

★ Extension of the Chapter 2 stock-flow relation. Not required — the graphical and two-period treatment is always sufficient.

4

The intertemporal model

Drills 9 and 10 — Chapter 3, the heart of the course

30 minutes

The model, in four assumptions

Notice how little is assumed. No firms, no government, no money, no exchange rate.

1

Two periods, and an endowment

The country receives Q_1 now and Q_2 later. It does not produce them with capital or labour — they simply arrive. It starts with a stock of foreign bonds B_0, which may be negative.

2

A budget constraint that holds over a lifetime

It can borrow or lend between the periods, but the present value of what it consumes must equal the present value of what it has. $B_2 = 0$: no debt is left unpaid and no wealth is left unspent.

3

Preferences that favour smoothness

Households want consumption in both periods and dislike having a lot of one and little of the other. This is the only behavioural assumption in the whole model, and it is what makes the current account move.

4

Free capital mobility

The country can borrow and lend at the world rate. Because it is small, it cannot affect that rate, so $r_1 = r^*$.

Four assumptions in. Three sharp, checkable predictions out. That is what a good model looks like.

The budget line, and the two things that move it

Get this right and half of Chapter 3 is free.

$$C_1 + C_2/(1+r^*) = (1+r_0)B_0 + Q_1 + Q_2/(1+r^*) \equiv \bar{Y}$$

$\bar{Y}$ is lifetime wealth: everything the country has, valued today.

Its slope is $-(1+r^*)$

Give up one unit of consumption today, put it in the bank, and you get $1+r^*$ units tomorrow. The slope is a price: the price of today's goods in terms of tomorrow's.

It always passes through the endowment point

The point (Q_1, Q_2) — consume exactly what arrives, borrow and lend nothing — is always affordable. Whatever happens to r^*, that point stays on the line.

A change in Q_1 or Q_2 shifts it

The slope does not change, because r^* did not change. The line moves out or in, parallel to itself, and the endowment point moves with it.

A change in r^* rotates it

The slope changes. But the endowment point is still affordable, so the line pivots about (Q_1, Q_2) — it does not shift.

Endowments shift the line. The interest rate rotates it. Mixing these up is the most common error on this material.

Drill 9

Three changes. For each one: shift or rotation, and which way?

9

A small open economy has endowments Q_1 and Q_2, starts with $B_0 = 0$, and faces a world interest rate r^*. Draw the budget line, mark the endowment point, and mark the optimum.

Now, on three separate diagrams, show what happens to the line when:

(a) Q_1 rises, with Q_2 and r^* unchanged;

(b) Q_2 rises, with Q_1 and r^* unchanged;

(c) r^* falls, with both endowments unchanged.

In each case say whether the line shifts or rotates, and name the point it moves around or through.

10 minutes

Label your axes C_1 and C_2, every time, without being asked. On the real paper this is worth marks.

While you work — a question to be ready for

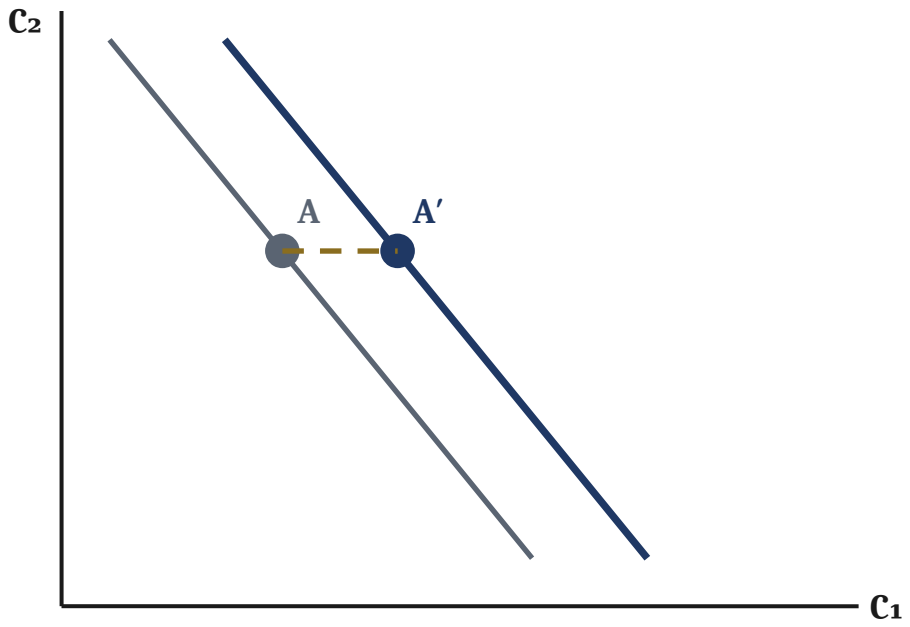
In (a) and (b) the country's lifetime wealth $\bar{Y}$ rises in both cases. Does it rise by the same amount? If not, which one raises it more, and why?

Chapter 3, slides 7 and 8.

Drill 9 — answers

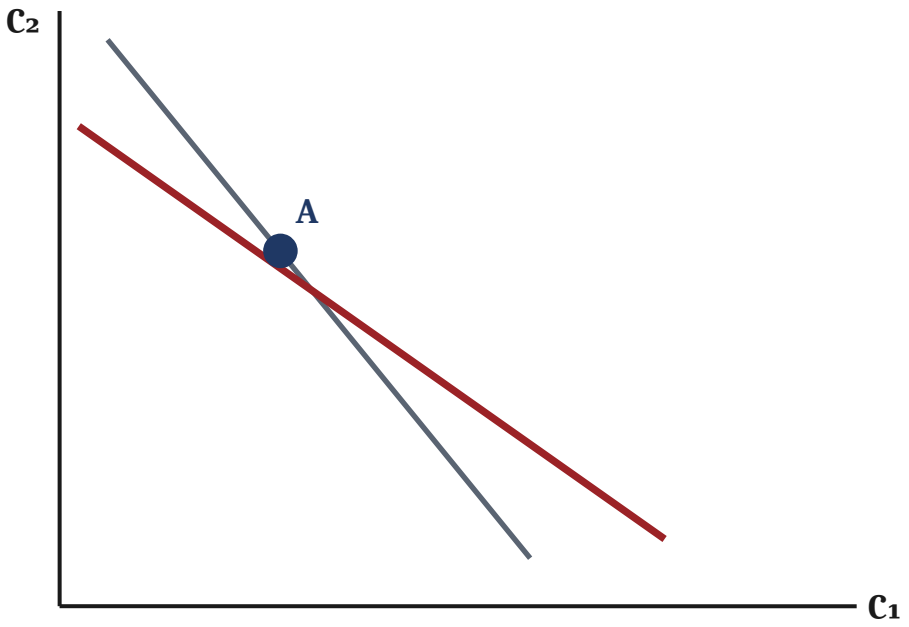
A shift and a rotation, side by side

(a) Q_1 rises



Parallel shift outward. The slope is still $-(1+r^*)$ because r^* did not move. The endowment point slides horizontally to the right, by exactly ΔQ_1.

(c) r^* falls

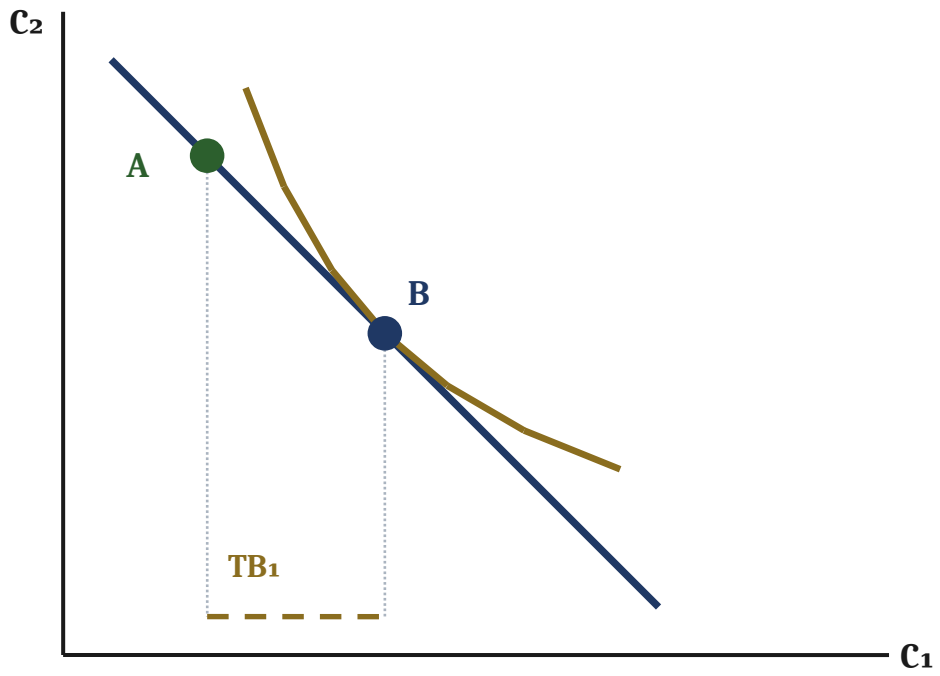


Rotation about the endowment point. The line gets flatter because the slope $-(1+r^*)$ is now smaller in magnitude. A does not move — consuming your own endowment is affordable at any interest rate.

(b) Q_2 rises: also a parallel shift, but the endowment point slides vertically — and $\bar{Y}$ rises by only $\Delta Q_2/(1+r^*)$, not ΔQ_2.

The optimum, and what the Euler equation is for

The tangency is the answer. The algebra just writes it down.



B is the optimum: the indifference curve just touches the budget line. A is the endowment. The gold gap between them is the trade balance — here a deficit, because the country consumes to the right of what it produces.

$$U'(C_1) = (1 + r^*) \cdot \beta \cdot U'(C_2)$$

The Euler equation. It is the tangency condition, written algebraically: the slope of the indifference curve equals the slope of the budget line.

What it means in words

At the optimum, giving up a little consumption today must cost exactly what it buys tomorrow. If it cost less, the household would save more; if more, it would save less. So there is nothing left to gain by moving.

What the exam needs from you

The graphical route is always sufficient: draw the line, draw a tangent indifference curve, mark B, and read off the trade balance as the horizontal distance between A and B. Deriving the Euler equation is starred — bonus credit, never required.

Two last pieces: why $r_1 = r^*$, and the formula that falls out

Free capital mobility, then log preferences

Why $r_1 = r^*$

Households can borrow and lend freely in world markets. If the domestic rate were above r^*, everyone would borrow abroad and lend at home until the gap closed. If it were below, the reverse. And because the economy is small, its own borrowing does not move r^*.

So the world rate is not an outcome of the model — it is an input the country has to take as given.

Then assume log preferences, $\beta = 1$

The Euler equation becomes $1/C_1 = (1+r^*)/C_2$, so $C_2 = (1+r^*)C_1$. Substitute that into the budget constraint and the whole thing collapses to one line:

$$C_1 = \tfrac{1}{2}\,\bar{Y}$$

Consume half your lifetime wealth in the first half of your life. Two periods, so one half each.

The four lines that solve any numerical question on this material

Lifetime wealth	$\bar{Y} = (1+r_0)B_0 + Q_1 + Q_2/(1+r^*)$
Consumption	$C_1 = \tfrac{1}{2}\bar{Y} \quad C_2 = (1+r^*)\cdot\tfrac{1}{2}\bar{Y}$
Trade balance	$TB_1 = Q_1 - C_1$
Current account	$CA_1 = TB_1 + r_0B_0 \quad \text{and} \quad B_1 = B_0 + CA_1$

Drill 10

The full numerical solution. This is question 3 on the mock paper.

10

A small open economy has logarithmic preferences and $\beta = 1$. It begins period 1 holding $B_0 = -20$ in foreign bonds — that is, it owes 20. The interest rate on that position was $r_0 = 10\%$, and the world rate is $r^* = 10\%$.
Its endowments are $Q_1 = 100$ and $Q_2 = 121$.

12 minutes

Find, in this order:

(a) lifetime wealth $\bar{Y}$

(b) consumption in both periods, C_1 and C_2

(c) the trade balance in period 1

(d) the current account in period 1 — and explain why it is not the same number as (c)

(e) the bond position the country carries into period 2, B_1

Then check your work: put your numbers into the period-2 budget constraint $C_2 + B_2 - B_1 = r_1 B_1 + Q_2$ with $B_2 = 0$, and see if it closes.

Chapter 3, slides 31 to 33.

Drill 10 — answers

Everything exact, and the period-2 constraint closes

(a)	$\bar{Y} = (1.1)(-20) + 100 + 121/1.1$	$-22 + 100 + 110 = 188$
(b)	$C_1 = \frac{1}{2}(188)$	94
	$C_2 = (1.1)(94)$	103.4
(c)	$TB_1 = Q_1 - C_1 = 100 - 94$	$+6$
(d)	$CA_1 = TB_1 + r_0 B_0 = 6 + (0.1)(-20)$	$+4$
(e)	$B_1 = B_0 + CA_1 = -20 + 4$	-16

The check

$C_2 + B_2 - B_1 = r_1 B_1 + Q_2$

Left: $103.4 + 0 + 16 = 119.4$

Right: $(0.1)(-16) + 121$
$= -1.6 + 121 = 119.4$

It closes exactly.

(d) — why the two numbers differ

The country sells 6 more goods than it buys, so its trade balance is +6. But it owes 20 from the past, and this year it pays 2 of interest on that. The current account is what is left: +4. A country can run a trade surplus and still see its external position improve by less than that surplus — the gap is yesterday's debt being serviced.

Note the direction of travel: $B_0 = -20$ becomes $B_1 = -16$. The surplus is paying the debt down, slowly.

10:00

Part 5 next: the three shocks. This is the most heavily examined part of the course.

5

The three shocks

Drills 11 and 12 — the most heavily examined material in the course

30 minutes

The three multipliers, on one slide

Learn this table. Then learn why each line is what it is.

Shock	What changes	ΔC_1	ΔCA_1	With $r^* = 10\%$
Temporary	$\Delta Q_1 > 0$, $\Delta Q_2 = 0$	$\frac{1}{2}\,\Delta Q_1$	$\frac{1}{2}\,\Delta Q_1$	$0.50 \times \Delta Q_1$
Permanent	$\Delta Q_1 = \Delta Q_2 > 0$	$\frac{1}{2}\left(1 + \frac{1}{1+r^*}\right)\Delta Q_1$	$\frac{1}{2} \cdot \frac{r^*}{1+r^*} \cdot \Delta Q_1$	$0.045 \times \Delta Q_1$
Anticipated	$\Delta Q_1 = 0$, $\Delta Q_2 > 0$	$\frac{\Delta Q_2}{2(1+r^*)}$	$-\frac{\Delta Q_2}{2(1+r^*)}$	$-0.45 \times \Delta Q_2$

The one line that explains all three

A unit of income today adds one unit to lifetime wealth. A unit of income next year adds only $1/(1+r^*)$. Households spend half of whatever wealth they end up with, whenever it arrives. Every number in the table above is that sentence, applied.

Note the signs. Good news about today improves the current account. Good news about tomorrow makes it worse. Nothing is wrong with the model — that is the prediction.

How to tell which shock you are looking at

Two questions, asked in order. Then you know which row of the table you need.

Question 1: does this year's output change?	Question 2: does next year's output change?
This year only	TEMPORARY
Both, equally	PERMANENT
Next year only	ANTICIPATED

TEMPORARY

The current account absorbs it. Consumption barely moves. The country borrows in a bad year and lends in a good one — exactly what a smoothing household does.

PERMANENT

Consumption absorbs it. The current account barely moves. There is nothing to smooth against: every future year is affected too, so borrowing only postpones the adjustment.

ANTICIPATED

The current account moves, in the opposite direction to the news. Consumption rises today on the strength of income that has not arrived yet, and that is financed by borrowing.

The lunch-money story, which is the whole chapter

Lose your lunch money one day and you borrow from a friend and pay them back next week: a temporary shock, financed, and your eating habits do not change. But if your father cuts your allowance for good, no friend can help — you have to eat less. That is a permanent shock, adjusted to. And if you learn you will get a raise next month, you might borrow a little now against it: anticipated good news, financed by going into deficit today.

Drill 11

Same country as drill 10. Three different pieces of news.

Baseline, from drill 10: $B_0 = -20$, $r_0 = r^* = 10\%$, $Q_1 = 100$, $Q_2 = 121$ → $\bar{Y} = 188$, $C_1 = 94$, $TB_1 = +6$, $CA_1 = +4$

11 Work out the new C_1, TB_1 and CA_1 in each of these three cases, taken separately, and in each case state the change in the current account:

(a) Q_1 rises by 11. Q_2 is unchanged.
(b) Q_1 and Q_2 both rise by 11.
(c) Q_2 rises by 11. Q_1 is unchanged.

Then check each answer against the multiplier formula from the table.

14 minutes

All three are increases in output. Before you compute: predict the sign of ΔCA_1 in each case, and write your prediction down.

The numbers are again exact. Nothing should have more than one decimal place.

Drill 11 — answers

One of these three is a deterioration

Case	C_1	TB_1	CA_1	ΔCA_1	From the formula
Baseline	94.0	+6.0	+4.0	—	—
(a) Temporary	99.5	+11.5	+9.5	**+5.5**	½(11) = 5.5
(b) Permanent	104.5	+6.5	+4.5	**+0.5**	½(0.1/1.1)(11) = 0.5
(c) Anticipated	99.0	+1.0	−1.0	**−5.0**	−11/(2 × 1.1) = −5.0

Read the middle row

Output rose by 11 in both periods. Consumption rose by 10.5 of it. The current account moved by half a unit. The country absorbed essentially the whole shock through consumption and barely touched the external accounts — which is what the model says it should do, because there is nothing to smooth against.

Read the bottom row

Good news — and the current account flips from a surplus of 4 to a deficit of 1. Nothing was produced, nothing was consumed abroad, no policy changed. The country simply learned that next year will be better, raised consumption today on the strength of it, and borrowed the difference.

Drill 12

Four scenarios. Name the shock, then predict the current account.

For each: which of the three shocks is it, which way does the current account move, and one sentence saying why.

(a) A country's grape harvest fails after a single bad summer. Meteorologists and growers both expect next year's harvest to be normal.

(b) A large offshore gas field is confirmed. It cannot produce for five years, but its size is public and not in doubt.

(c) The national statistical office revises its estimate of the country's long-run productivity growth down, and the revision applies to every year ahead.

(d) Imported energy prices spike sharply. The central bank and the Commission both publish forecasts in which prices unwind within two years.

10 minutes

Two of these four are the same kind of shock. Say which two, and what makes them the same despite looking nothing alike.

Notice what the model does not need to know: the cause. A bad harvest and an energy-price spike are the same object to this model, and that is a feature, not a gap.

Drill 12 — answers

Two temporary, one anticipated, one permanent

(a)	**TEMPORARY, negative**	**CA worsens**	Q_1 falls, Q_2 does not. The country borrows to keep consumption up and the deficit absorbs the shock.
(b)	**ANTICIPATED, positive**	**CA worsens**	Q_2 rises, Q_1 unchanged. Consumption rises now against income that has not arrived, financed by borrowing.
(c)	**PERMANENT, negative**	**CA barely moves**	Q_1 and Q_2 both fall. Consumption falls by nearly the whole amount; there is no good year to smooth against.
(d)	**TEMPORARY, negative**	**CA worsens**	Same object as (a). Only the size and the headline differ.

Why (a) and (d) are the same shock

Because the only thing the model asks about a shock is which periods it hits. A failed harvest and an energy-price spike arrive through completely different channels, and both show up here as: Q_1 down, Q_2 unchanged. The model gives them the same answer, and the data broadly agrees — which is the sort of thing a model earns its keep by doing.

The same three shocks, in the euro area

What the model predicted, and what happened

Spain, 1999–2007

Anticipated

Joining the euro brought interest rates down and a decade of catch-up growth was widely expected. Both effects push the same way: borrow now against a richer future.

What happened

A current-account deficit that widened year after year, reaching roughly a tenth of GDP.

Verdict

The model gets the direction and the persistence right.

Greece, 2008–2015

Future income revised down

The expected path of income was marked down sharply and repeatedly. Households and the state had to adjust rather than finance.

What happened

A deficit of over 10% of GDP swung into surplus — a reversal of roughly 14 points.

Verdict

Direction right, and the model under-predicts the size. The missing piece is the sudden stop: lending did not merely reprice, it stopped.

EU, 2022

Temporary

Energy prices spiked and forecasters expected them to unwind. The model says: finance it, do not adjust to it.

What happened

The EU current-account surplus very nearly vanished in 2022, then recovered to a record by 2024.

Verdict

A completed natural experiment, and close to the textbook answer.

An honest verdict: two fits and one near miss. The near miss is informative, and saying so is worth marks.

Figures as presented in the Session 5 slides. Sources: Eurostat and ECB balance of payments.

6

The mock paper

One hour, sixty marks, four questions

15 minutes

Mock paper — questions 1 and 2

24 marks, 24 minutes. Full solutions in the online practice set.

Question 1 — accounting

12 marks

A country reports, in billions of euro: output 240, consumption 150, investment 55, government spending 48. It also pays 7 of net investment income to foreigners.

(a) Find absorption and net exports. (b) Find the current account. (c) Find national saving, and verify that CA = S − I. (d) In one sentence, say why the trade balance and the current account differ here.

Question 2 — stocks, flows and valuation

12 marks

The same country begins the year with a net international investment position of −320. Over the year its foreign assets gain 18 in value, and foreign holdings of its domestic assets gain 31.

(a) Find the net valuation change, and the position at the end of the year. (b) How much of the change in the position did the country choose? (c) Next year it halves the current-account deficit. Does its net external debt fall? Explain.

Question 2 uses your answer to question 1(b). If you could not get it, assume a deficit of 20 and carry on — on the real paper, say so and you keep the marks.

Mock paper — questions 3 and 4

36 marks, 36 minutes, plus the starred bonus

Question 3 — the endowment economy

24 marks

A small open economy has logarithmic preferences and $\beta = 1$. It holds $B_0 = +40$ of foreign bonds, $r_0 = r^* = 25\%$, and endowments $Q_1 = 80$, $Q_2 = 150$.

(a) Find $\bar{Y}$, C_1, C_2, TB_1, CA_1 and B_1. (b) Draw the budget line, the endowment point and the optimum, and mark the trade balance on your diagram.
(c) Now Q_1 rises by 20, with Q_2 unchanged. Find the new CA_1 and check it against the multiplier. (d) Instead, Q_2 rises by 25, with Q_1 unchanged. Find the new CA_1.
(e) Compare your answers to (c) and (d). Something surprising has happened to consumption. Say what, and explain it.

Question 4 — name the shock

12 marks

For each, name the shock, give the direction of the current account, and justify in one sentence.
(a) A drought cuts this year's output; rainfall is expected to be normal next year. (b) A country is told its EU structural funds will double from next year onwards. (c) A long-running industry closes permanently and will not be replaced.

Starred bonus ★ up to 6 marks, never required

Starting from $C_1 = \tfrac{1}{2}\bar{Y}$, derive the permanent-shock multiplier $\Delta CA_1 = \tfrac{1}{2} \cdot r^*/(1+r^*) \cdot \Delta Q_1$, and say what it tends to as $r^* \to 0$.

How to spend the hour

Sixty marks, sixty minutes. One mark, one minute.

First 3 minutes	**Write out the four equations**	Before you read anything. Wealth, consumption, trade balance, current account — and the stock-flow relation. You now have a reference sheet in your own handwriting.
Minutes 3 to 15	**Question 1**	Accounting. No model, no diagram, nothing to think about. Bank these marks and do not linger.
Minutes 15 to 27	**Question 2**	Stocks and flows. Watch the sign on the liability valuation. Two of the three parts are sentences, not sums.
Minutes 27 to 51	**Question 3**	The big one. Follow the four lines in order. Draw the diagram even if you do not need it — it carries its own marks.
Minutes 51 to 60	**Question 4**	Three labels and three sentences. Quick marks, which is why it is last and why you must leave time for it.

The graphical and intuitive route is sufficient for every question on the paper. Nobody has to derive anything to score full marks on the sixty.

7

The next two weeks

What to actually do between now and 27 November

10 minutes

A plan for the fortnight

Two weeks is plenty, if you spend it on the right things

This week	**Redo today's twelve drills, from a blank page** Not reading them — doing them. If you cannot reproduce drill 10 without looking, that is exactly the gap the exam will find.
This week	**Write out the four equations until you can do it in two minutes** This is the only pure memorisation the course asks of you. It takes a few attempts and then it is permanent.
Next week	**Sit the mock paper in one hour, with no notes and a clock** Timed, once, properly. Then mark it against the solutions. A mock you take at leisure tells you nothing about an exam you take under pressure.
Next week	**Work the online practice set** Thirty questions with full worked solutions, and the mock paper with its mark scheme. Reveal the answer only after you have written something.
The day before	**Reread the three summary slides, and stop** The scope slide, the three-multiplier table, and the decision rule. Nothing new the day before an exam — it only shakes your confidence in what you already know.

Everything above is on the course page: the practice set, the mock paper and the solutions.

What you can now do

Read a current account

From either side of the identity, with or without income from abroad, and refuse to moralise about the sign.

Tell a deficit from a debt

And explain why cutting one does not undo the other — the single most useful thing in the course outside an exam hall.

Solve the model

Lifetime wealth, consumption, trade balance, current account. Four lines, in order, and a check that closes.

Predict and then explain

Which shock, which direction, and why — including the case where good news makes the current account worse.

Friday 27 November · one hour · the graphical route is always sufficient